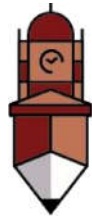


The University of Faisalabad

Department of Computer Sciences



The
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Faisalabad

Computer Networks
CS-314

This is a sample for your help. Please do not use it in your lab manual.

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Lab No. 01: Introduction to Cisco Packet Tracer

Learning Objectives / Outcomes:

After completion of this Lab, students will be able:

- To understand the usage and need of tool cisco packet tracer.

Lab Work:

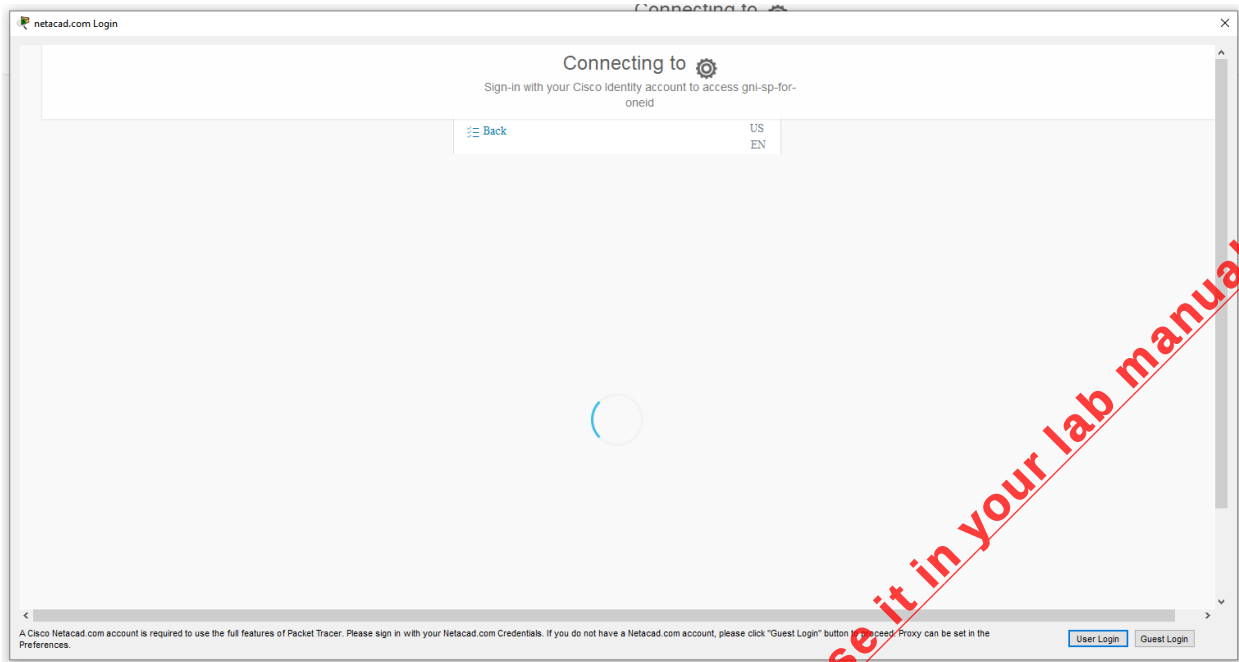
Cisco Packet Tracer as the name suggests, is a tool built by Cisco. This tool provides a network simulation to practice simple and complex networks.

Main purpose:

The main purpose of Cisco Packet Tracer is to help students learn the principles of networking with hands-on experience as well as develop Cisco technology specific skills. Since the protocols are implemented in software only method, this tool cannot replace the hardware Routers or Switches. Interestingly, this tool does not only include Cisco products but also many more networking devices.

Cisco Packet Login

This is a sample for your help. Please do not use it in your lab manual.



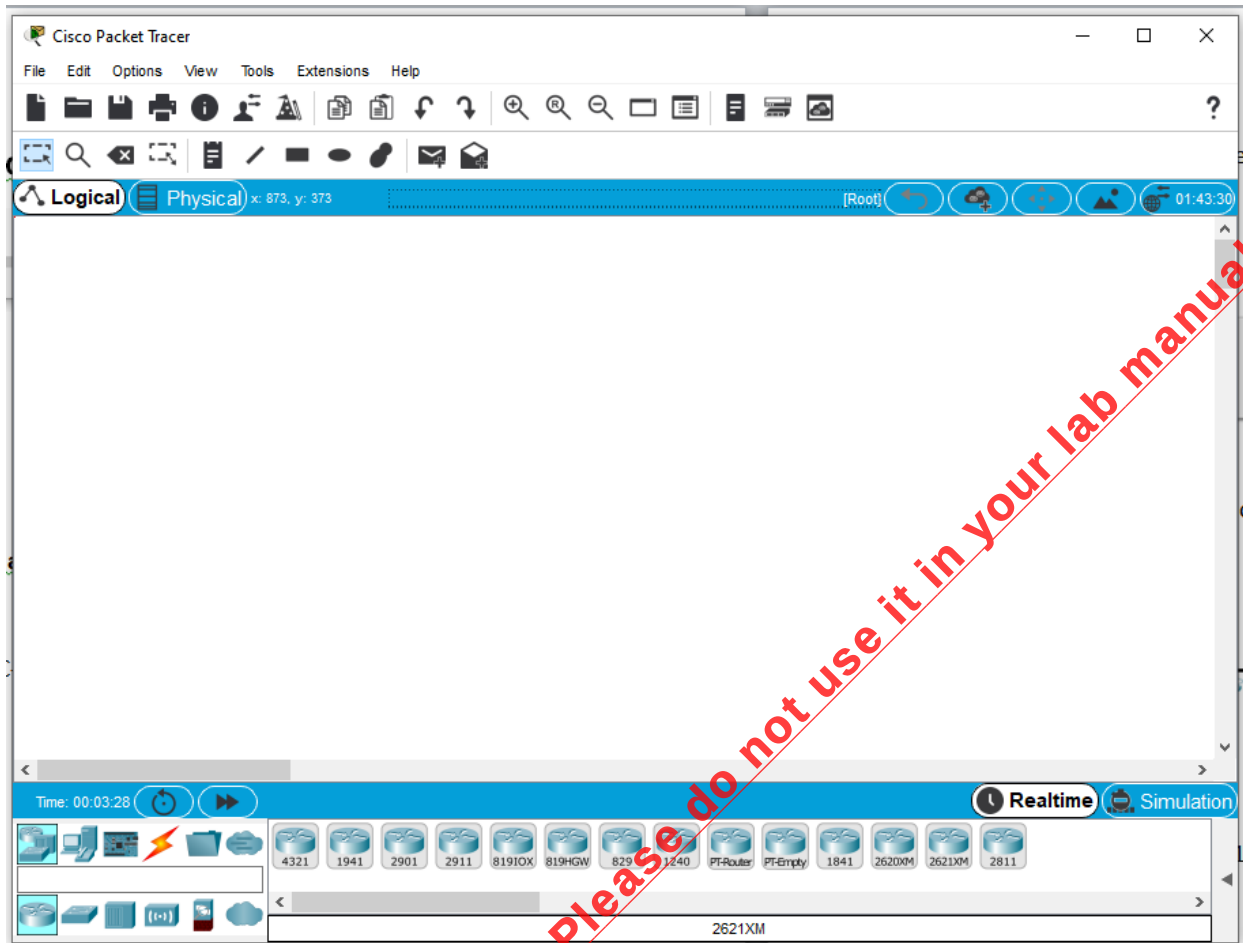
Login as a guest:



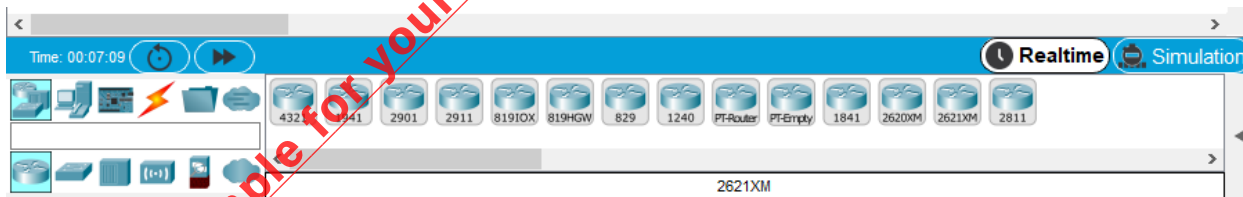
After few seconds confirm as a guest.



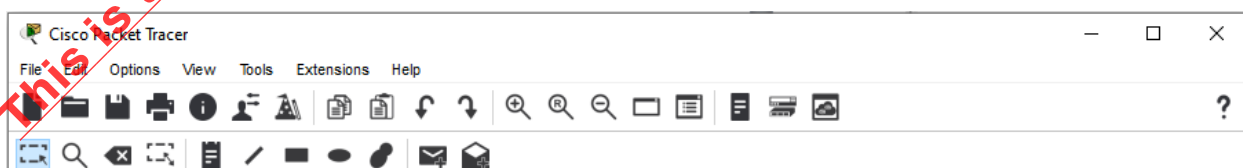
Internal view of cisco packet Tracer:



Use switch, router, pc, end devices etc. from:



Use notes message simulation etc. from:



Lab No. 02: IP Assigning, Commands for CMD, Labeling method, Simple packet Transformation

Learning Objectives / Outcomes:

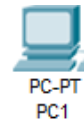
After completion of this Lab, students will be able:

- To understand the concepts of IP assigning and commands for cmd.
- To understand the Labeling method and Simple packet Transformation

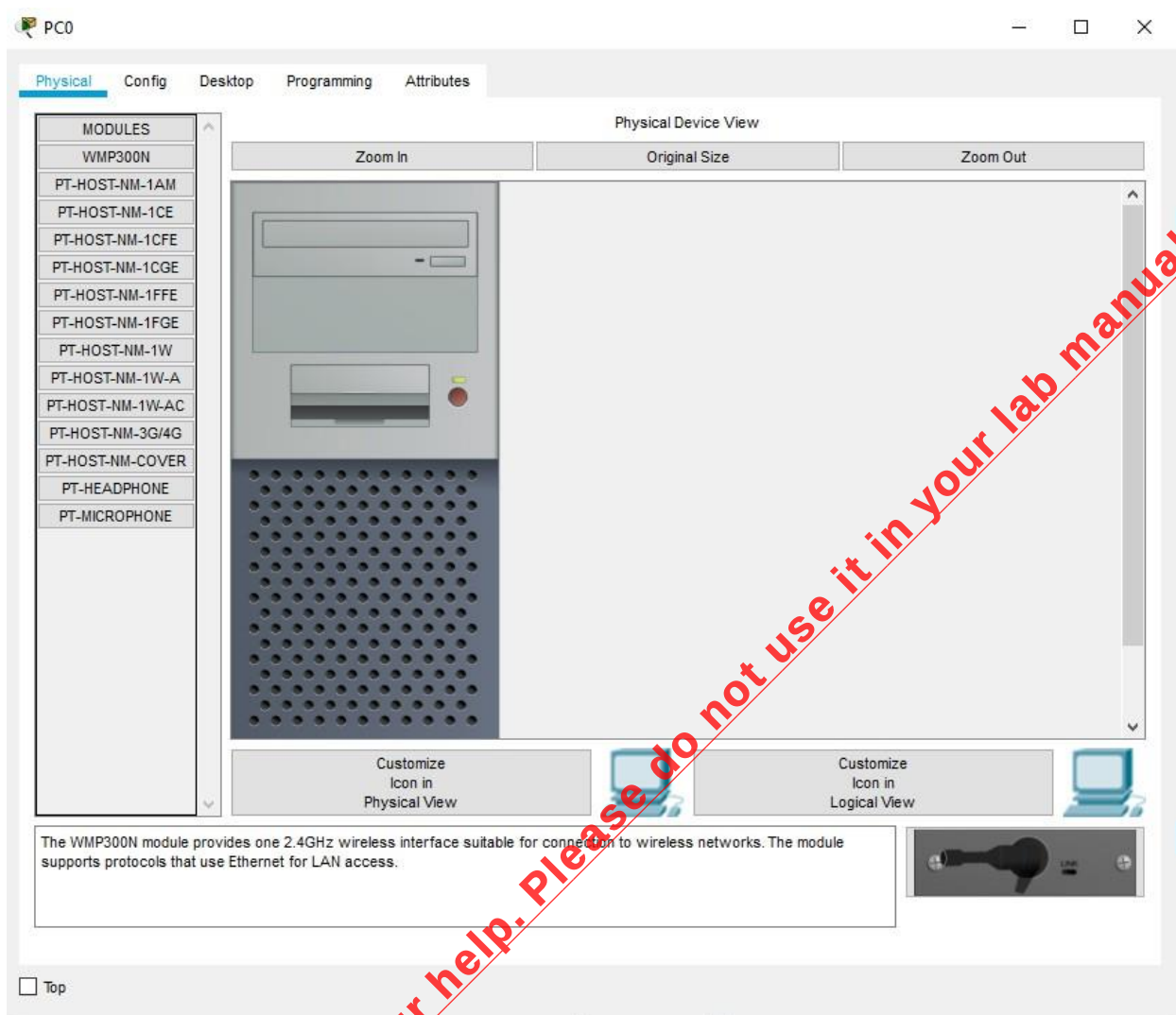
Lab Work:

1- IP Assigning:

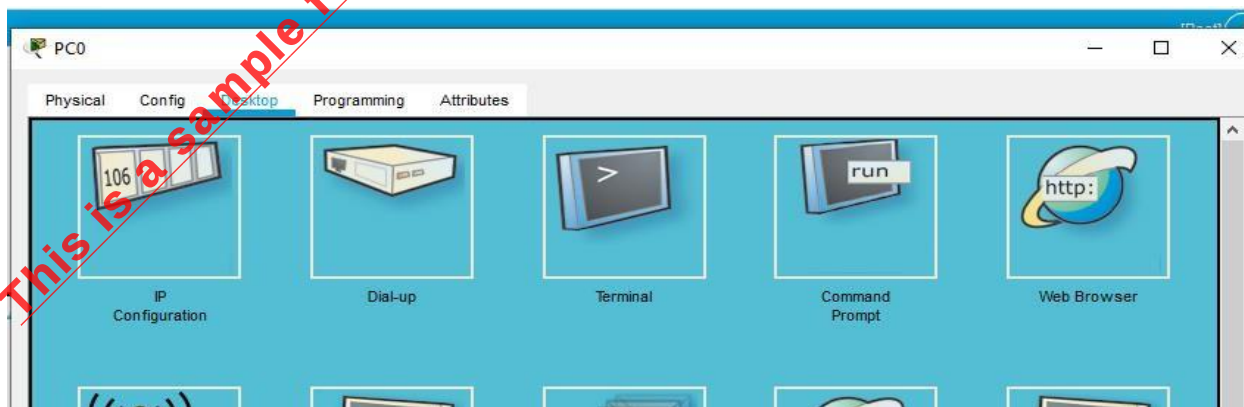
Take PC-0 and PC-1 from End devices.



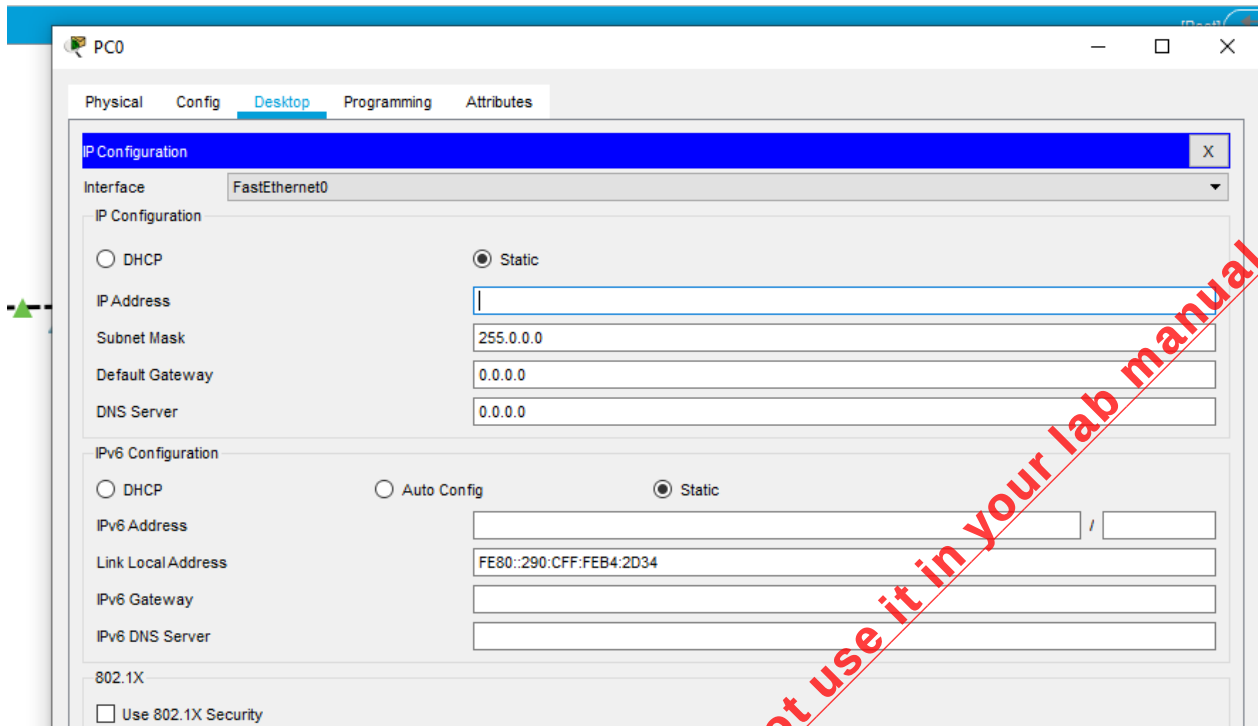
Now, Make Logical connection between both computers.
Click on PC-0 dialogue box will appear



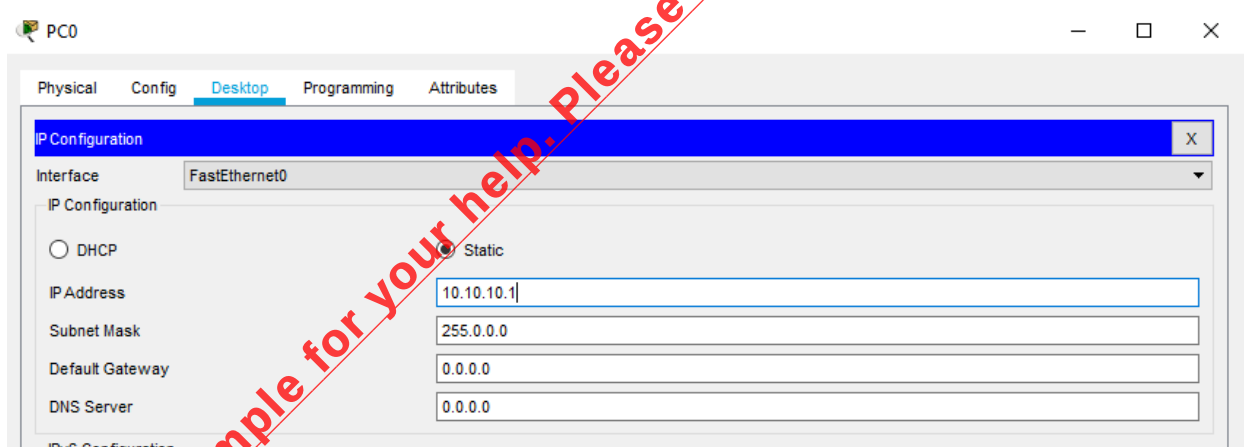
Select desktop from Header, desktop option appear.



Select IP configure and assign IP Address to PC-0



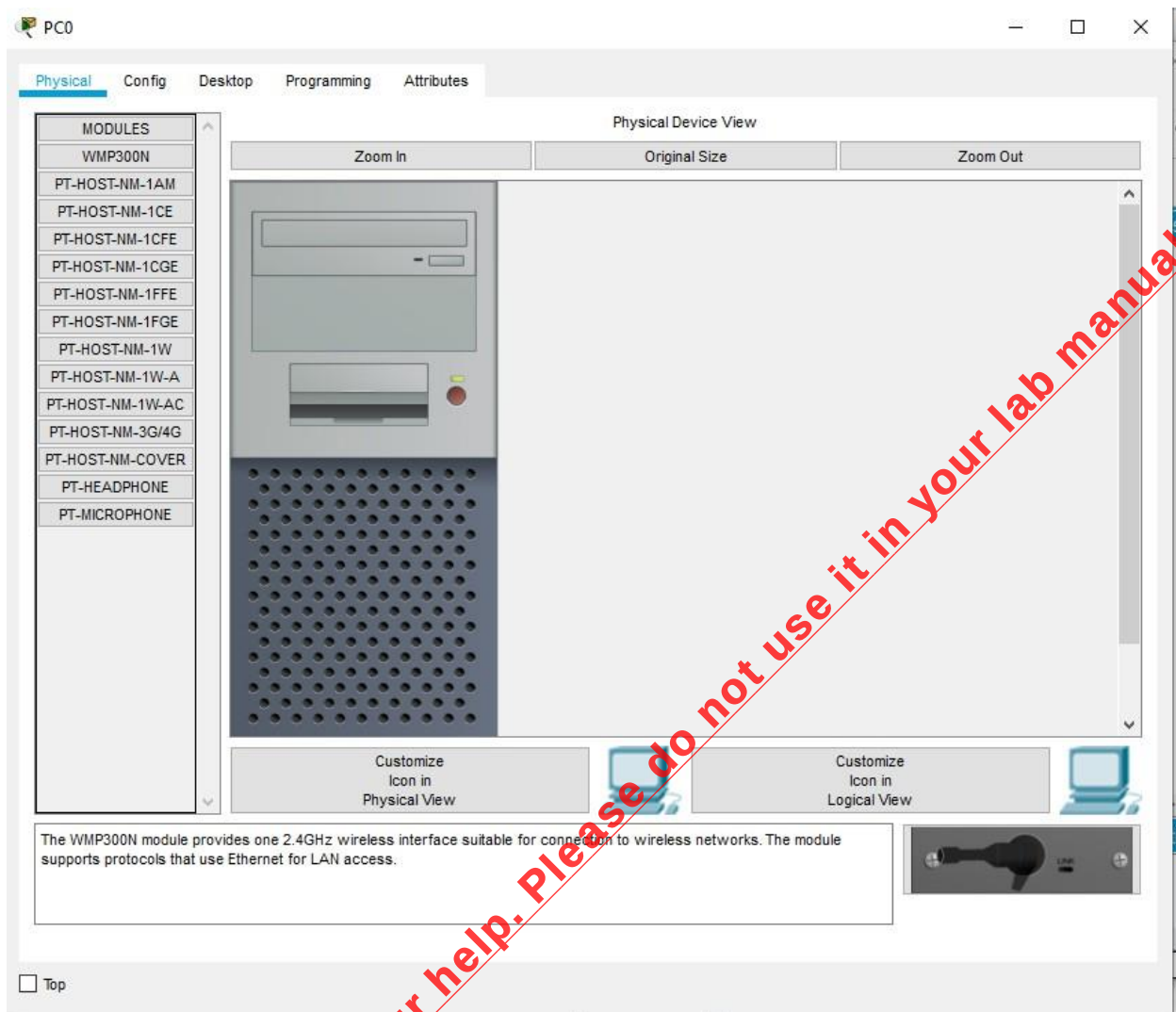
Assign IP address.



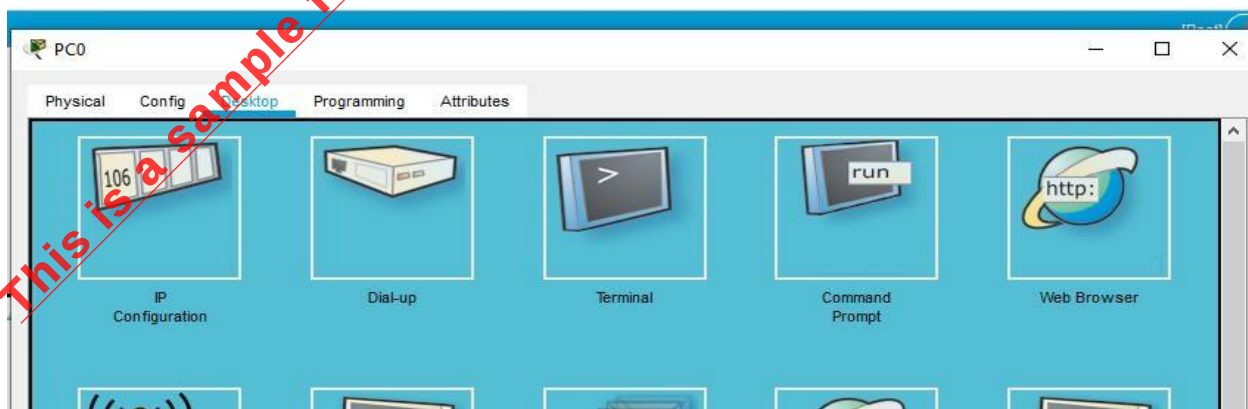
Similar with PC-1.

2- Commands for CMD:

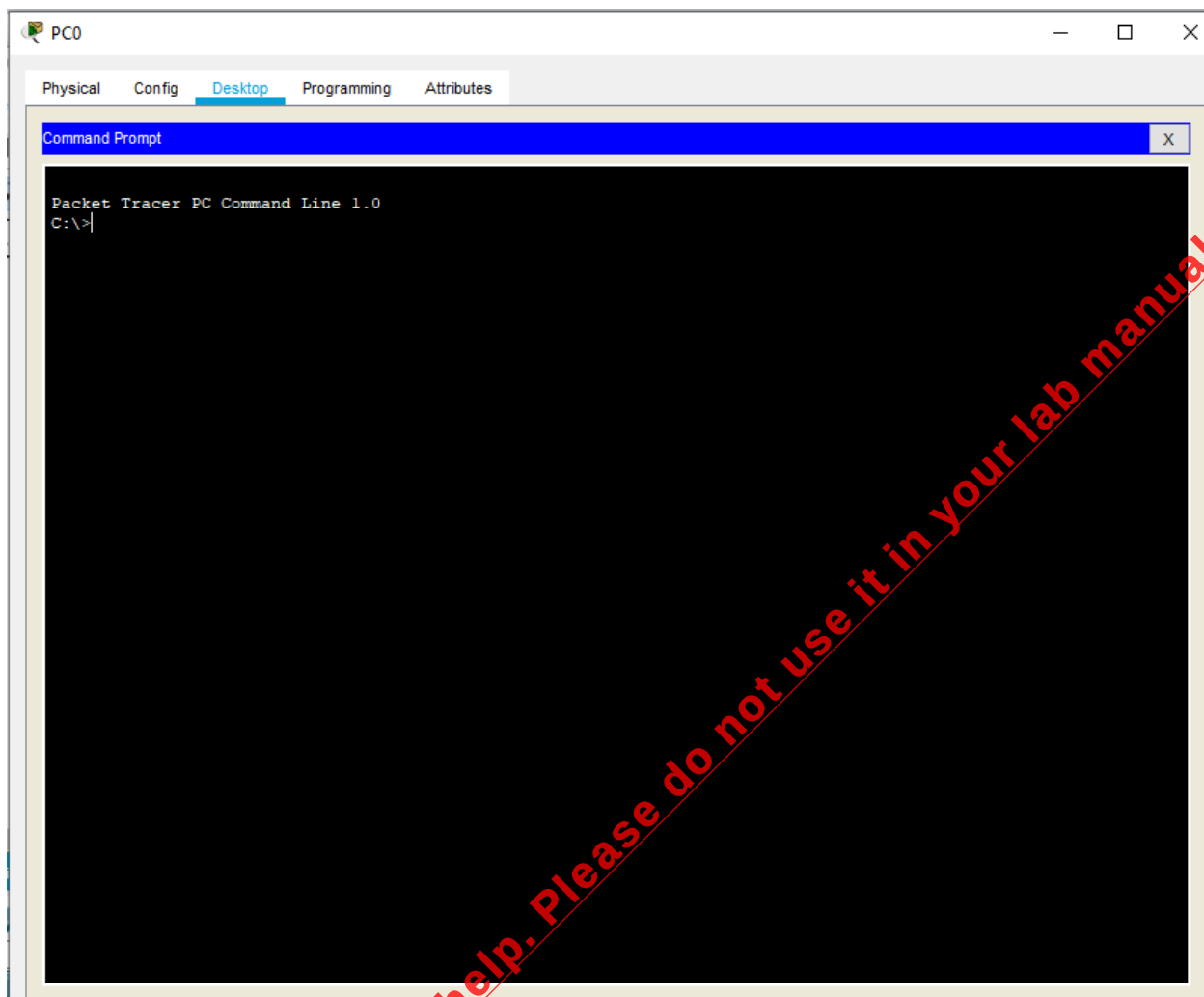
Check connect, click PC-0 dialogue box will appear.



Select desktop from Header, desktop option appear.



Click on command prompt.



Write ping command with PC-1 IP address.

```
C:\>
C:\>ping 10.10.10.2

Pinging 10.10.10.2 with 32 bytes of data:

Reply from 10.10.10.2: bytes=32 time=1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time<1ms TTL=128
Reply from 10.10.10.2: bytes=32 time=3ms TTL=128

Ping statistics for 10.10.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>
```

For check IP Address of PC-0. Use **ipconfig /all** command

```

C:\>ipconfig /all

FastEthernet0 Connection: (default port)

    Connection-specific DNS Suffix...:
    Physical Address.....: 0090.0CB4.2D34
    Link-local IPv6 Address.....: FE80::290:CFF:FEB4:2D34
    IP Address.....: 10.10.10.1
    Subnet Mask.....: 255.0.0.0
    Default Gateway.....: 0.0.0.0
    DNS Servers.....: 0.0.0.0
    DHCP Servers.....: 0.0.0.0
    DHCPv6 Client DUID.....: 00-01-00-01-73-D1-58-C0-00-90-0C-B4-2D-34

Bluetooth Connection:

    Connection-specific DNS Suffix...:
    Physical Address.....: 0000.0C78.99C7
    Link-local IPv6 Address.....: ::
    IP Address.....: 0.0.0.0
    Subnet Mask.....: 0.0.0.0
    Default Gateway.....: 0.0.0.0
    DNS Servers.....: 0.0.0.0
    DHCP Servers.....: 0.0.0.0
    DHCPv6 Client DUID.....: 00-01-00-01-73-D1-58-C0-00-90-0C-B4-2D-34

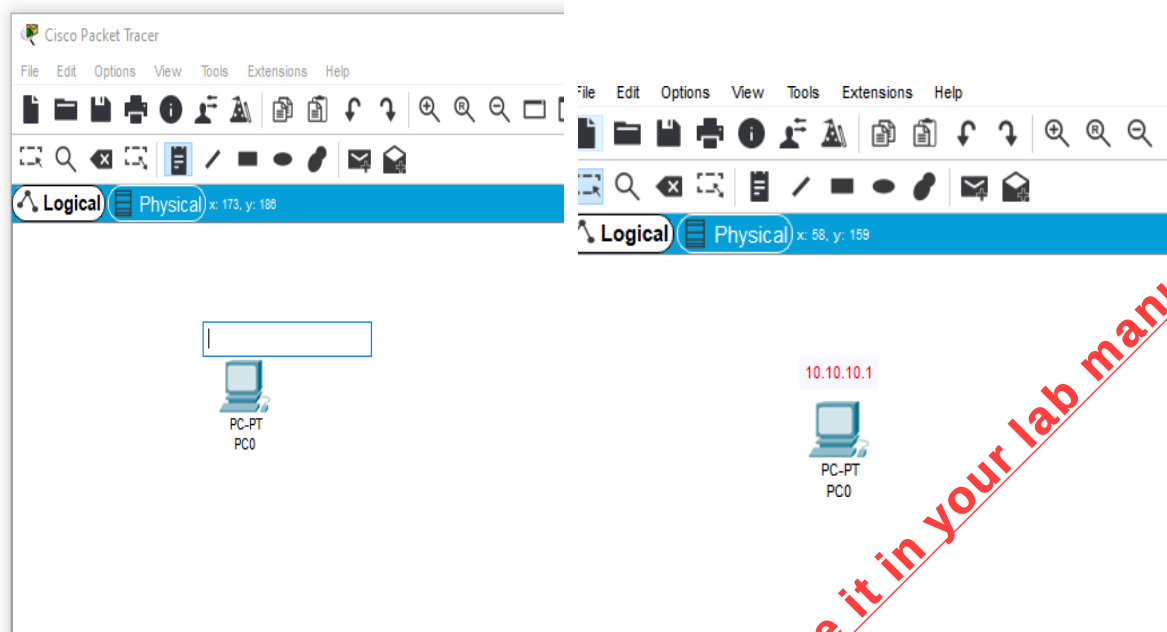
```

3- Labeling Methods:

Select place note from header.

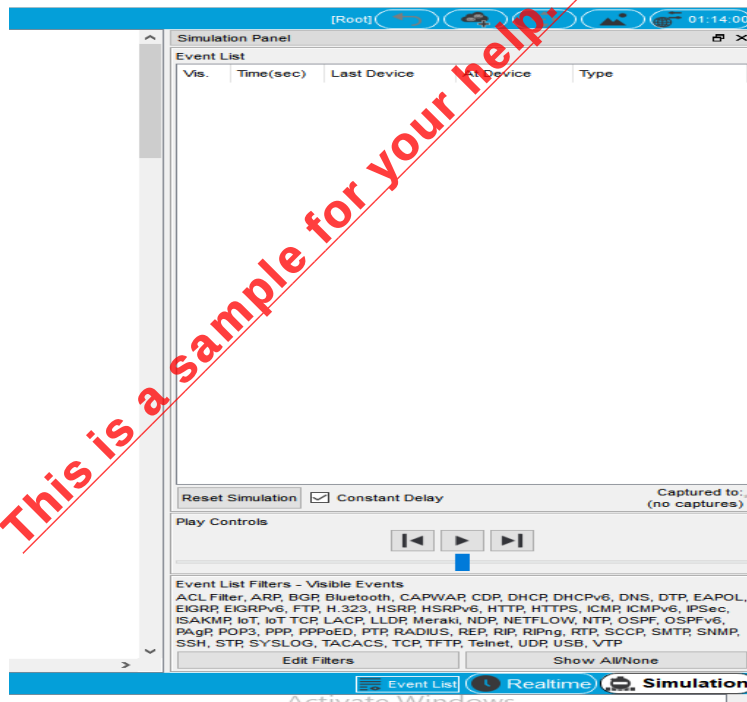


Click on screen and write note.

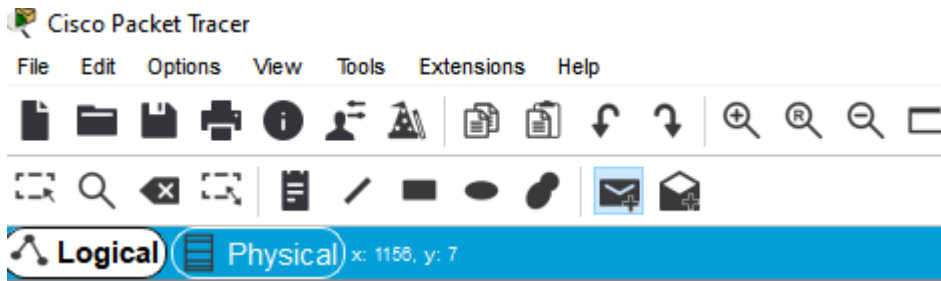


4- Simple packet Transmission:

Turn on your Simulation.



Select message from header.



Assign message.



Message transmission:

